



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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Computer Fundamental

Input/Output Device

Perioheral Device → Peripheral devices are external devices connected to a core system of computer to provide input,output or additional storage.

Example – Keyboard, printer, HDD,.....

❖ **Input Device**

The Input devices are the devices which are used to enter the data in the computer system.

e.g. - Keyboard, mouse, scanner, microphone are the example of input devices.

➤ **FUNCTIONS OF INPUT DEVICES:**

- Accept the data from the outside worlds.
- Convert that data into computer coded information.
- Supply this data to Central Processing Unit for further processing.

i. Keyboard

- Keyboard is most commonly used input device.
- It is similar like a type writer which is used to enter data in the computer.
- It contains sets of keys such as alphabets, number & special signs

✓ **Two types of keyboard**

- i. General purpose keyboard

ii. Special purpose keyboard

➤ General purpose keyboard

- This type of keyboard is a Standard keyboard which are used in personal computers.
- Most popular general purpose keyboard contains 101 keys.



➤ Special Purpose Keyboard

- Special purpose keyboard is used for special purpose applications which required faster data entry and rapid interaction with the computer system.
- For example ATM used in banks used special purpose keyboard which contains a few keys.



ii. Mouse

- Mouse is Small hand-hold Input device which is generally used for drawing purpose.
- It's a Pointing device.
- It contains two or three buttons.
- Left button is used to point out or select any item by clicking.
- Right to generate context menu.
- Many types of mouse are available such as mechanical mouse, optical mouse, serial mouse, wireless mouse which are used for different purpose.



iii. JOYSTICK

- Joystick is a pointing device which works on the same principle of track ball.
- It contains a stick which is placed on the spherical ball.
- The stick is used to move the cursor at desired position left or right or backward or forward.
- Some of the systems using joysticks are - Aircrafts, UAVs for flight control, Motorized Wheelchairs as input device, Microscopes, Submarines, Security Systems, Video Games.
- Joysticks are widely used for video games



iv. Track Ball

- Trackball is a pointing device which is similar to a mouse.
- In case of button press finger to up or down & left or right to move cursor.



v. Light Pen

- Light pen is a pointing device which is used to draw directly on the screen.
- It is called light pen because it is similar to a pen & senses light.
- The light pen allows the user to point out or draw any object on the screen.
- It is useful for drawing or graphics in the program such as CAD (computer aided design).
- An engineer, architect or fashion designer can draw directly on screen.
- Used in application such as gaming, graphic arts, healthcare applications etc.
- Light pen cannot scratch or damage a screen.



vi. Touch Screen

- Touch screen is a pointing device.
- It is most simple & easiest to learn of all input devices.
- A touch screen is an electronic visual display that can detect the presence and location of a touch within the display area.
- Touch screen are commonly used in following places.

a) ATM	d) Mobile
b) BANK	e) Air-Line
c) Super Market	f) Railway Ticket Machine...



vii. DIGITIZER (डिजिटाइजर)

- An input device.
- Used for converting pictures, map & drawing into digital form.
- Commonly used in CAD by architects & engineers.



viii. MICROPHONE

- It's an input device.
- Used to store the voice data into the computer system.
- Microphones are a type of transducer - a device which converts energy from one form to another.
- Microphones convert sound waves into electrical energy.
- All the microphones share one common thing: The diaphragm.
- When the diaphragm vibrates, it causes other components in the microphone to vibrate.
- These vibrations are converted into an electrical current which

becomes the audio signal.



ix. Web Camera

- It's an input device.
- Web camera is a hardware camera connected to a computer that allows everyone to connect to internet to view either pictures or motion video.
- Most Web cameras are embedded to display with laptop computer or connected with USB or Wi-Fi with a computer.



x. SCANNER

- Scanners are input devices.
- They are capable of entering information directly into the computer.
- The main advantage of direct entry of information is that users do not have to key the information.
- And another advantage is that through Scanners you can input Graphical Data into the computer.

- Provides faster and more accurate data entry. Important types of scanners

Types of Scanner

- Image Scanners – image, documents, text,
- Optical Scanners

✓ OMR (OPTICAL MARK READER)

OMR is a device that is capable of recognised pre-specified type of mark made by pencil or pen.

✓ OCR (OPTICAL CHARACTER READER) DEVICE

OCR capable of recognizing alphabets & numbers printed on paper. The type of document must be type using OCR fonts.

✓ MICR (MAGNETIC INK CHARACTER RECOGNITION)

MICR systems use special ink which can be magnetized, to print characters that can then be read and decoded by special magnetic devices. e.g. – Bank Cheques.

✓ BCR (BAR CODE READER) DEVICE

Data can be coded in the form of small lines which are known as Bar Codes. It uses laser-beam technology. Laser stroke across the pattern of bar which sensed by light sensitive decoder & their reflection of light pattern are converted into electrical pulse which converted it into alphanumeric value.



xi. Bio Metric Sensor

A biometric sensor is an input device that detects unique physical or behavioral characteristics of a human.



❖ OUTPUT Device

The output devices are the devices which are used to display the result generated by the computer system.

e.g. - Monitor, printer, plotter, speaker are the example of output devices.

✓ FUNCTIONS OF OUTPUT DEVICES:

- Accept the result from the CPU.
- Convert that result into human readable form(Binary number → Binary Code).
- Supply this result to output device.

i. Monitor

- The monitor is the common output device mostly used It is a softcopy output device.
- A monitor device that displays visual information such as text, images, and videos generated by the computer.
- The monitor can also determine basis of technology, color capability, size, resolution, purpose and graphical objects (pictures).

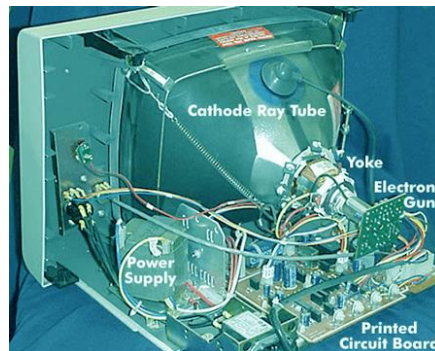


✓ **Two types of monitors are used.**

- i) CRT monitors. (CATHOD RAY TUBE)
- ii) Non CRT monitors (4 Types)
 - a) LCD (Liquif Crystal Display)
 - b) LED (Light Emitting Diode)
 - c) OLED (Organic Light Emitting Diode)
 - d) Plasma Display

i) CRT monitors

- In CRT display CRT is a specialised vacuum tube in which images are produced when electron beam strikes a phosphor surface.
- CRT monitor contains various electric parts like - cathode (filament), control grid (control the brightness), acceleration anode (focusing lens), deflection plates & phosphor coated screen.



ii) Non CRT monitors

a) LCD (Liquid Crystal Display)

- An LCD monitor uses liquid crystal material to control light and display images. It requires a backlight to produce images.
- Thin, lightweight and Low power consumption.

b) LED (Light Emitting Diode)

- An LED monitor is an advanced form of LCD monitor that uses LED backlighting instead of fluorescent lamps(ऐसी lamp जो gas और fluorescent coating की मदद से रोशनी पैदा करती है।).
- Brighter and more energy-efficient than LCD.
- Slimmer design and better contrast and longer life.



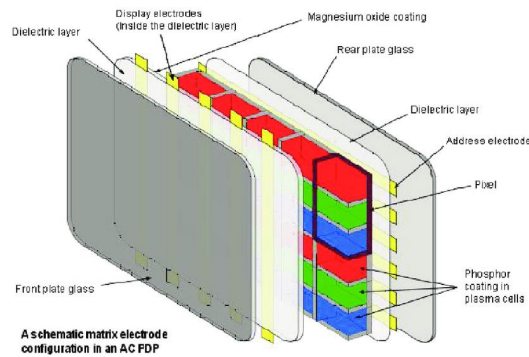
c) OLED (organic Light Emitting Diode)

- An OLED monitor uses organic compounds that emit light individually when electric current passes through them, so no backlight is needed.
- Very high contrast ratio, Deep black and vivid colors and very thin and flexible displays.



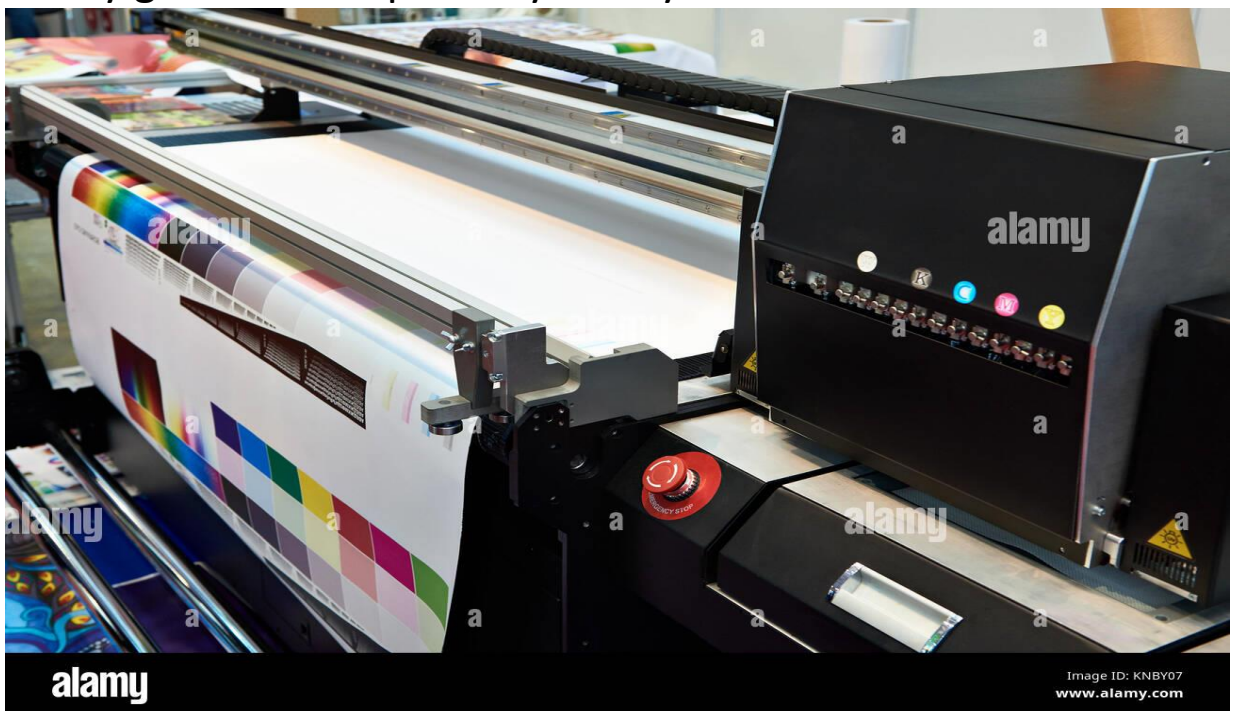
d) PDP (Plasma Displays Panel)

- A Plasma monitor uses small cells filled with ionized gas (plasma) that glow when electrically charged to produce images.



ii. PLOTTER

- Plotter is an output device which is capable to producing hardcopy output of graphics.
- Used for printing = Vector graphics
- Two type of PLOTTER = Drum & Flatbed
- Used to producing wide format printing.
- Plotters are used to print out large/huge graphs, designs, and other types of paper-based documents, such as engineering drawings, business charts, construction maps, and architectural blueprints generate hardcopy output of widely large in sizes.
- Normally generate output very slowly.



iii. SPEAKER

- The speaker is output device which is connected to computer's

soundcard.

- The speaker output the sound generated by the sound card.
- It can translate data into audio signal which are sending to speaker.



iv. PROJECTOR

- A projector is an output device that presents images onto a projection screen and can be connected to a computer, and can replace a monitor or television to project images to a big group of people.
- Projectors come in various forms and sizes and are used in auditoriums and movie theatres.





V. Braille Display

- A Braille Display is a special output device used by visually impaired (blind) users to read computer output through touch instead of sight.
- It works by converting text on the screen into Braille characters, which are formed by raised dots that can be felt with the fingers.

How to Work

- Displays information in Braille format using raised pins.
- Allows blind users to read text by touch.
- Connects to a computer or mobile device via USB or Bluetooth.
- Updates dynamically as the screen content changes.



VI. VR Headset

- A VR Headset (Virtual Reality Headset) is a head-mounted output

device that creates a computer-generated, immersive 3D virtual environment for the user.

- It allows the user to see, hear, and sometimes interact with a simulated world as if they were physically present inside it.



VII. Printer

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